AP CALCULUS AB

Unit 1 • Worksheet 1-7A • Answer key

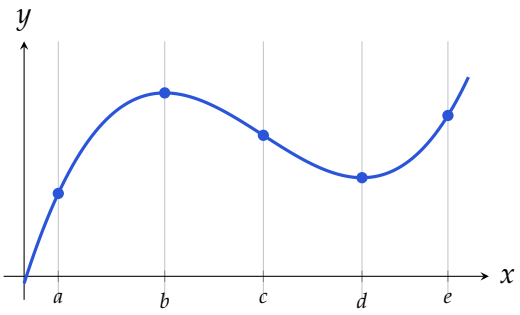
Worksheet 1-7A • Instantaneous Rates of Change

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Questions 1 to 3 refer to the graph of f.



Question 1

NO CALCULATOR

At which labeled point is $f'(x) < 0$?

(A) a

(B) b

(C) c

(D) e

Solution

Answer: (C). The graph falls at c.

Question 2

NO CALCULATOR

At which two labeled points is $f'(x) = 0$?

(A) b and d

(B) a and e

(C) b and c

(D) c and d

Solution

Answer: (A). The peak b and the valley d have horizontal tangent lines.

Question 3

NO CALCULATOR

Which statement about $f'(a)$ and $f'(e)$ is true?

(A) Both are negative.
(B) $f'(a) > 0$ and $f'(e) < 0$.
(C) Both are zero.
(D) Both are positive.

Solution
Answer: (D). The graph rises at a and at e.

Question 4

NO CALCULATOR

$\frac{dy}{dx}$ at $x = c$ means

(A) the average rate of change on $[0, c]$
(B) the slope of the tangent line at $x = c$
(C) the value $y(c)$
(D) the area under the graph

Solution
Answer: (B). Instantaneous rate of change = derivative = slope of the tangent line.

Question 5

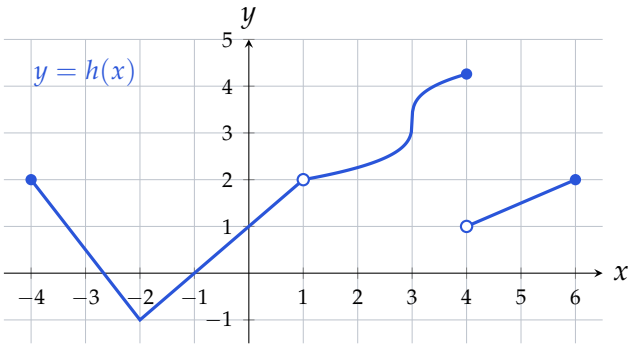
NO CALCULATOR

Where is $f(x) = |x + 1|$ not differentiable?

(A) $x = 1$
(B) $x = 0$
(C) $x = -1$
(D) nowhere

Solution
Answer: (C). Sharp corner at $x = -1$.

Questions 6 and 7 refer to the graph of h on $[-4, 6]$.



Question 6

NO CALCULATOR

At which x does h have a sharp corner?

(A) $x = -2$
(B) $x = 1$
(C) $x = 3$
(D) $x = 4$

Solution
Answer: (A). Two straight pieces with different slopes meet at $x = -2$.

Question 7

NO CALCULATOR

Why is h not differentiable at $x = 3$?

(A) It has a hole there.
(B) It has a jump there.
(C) It has a sharp corner there.
(D) It has a vertical tangent line there.

Solution
Answer: (D). The graph is continuous but becomes vertical at $x = 3$.

Question 8

NO CALCULATOR

Which statement is true?

(A) Continuous implies differentiable.
(B) Differentiable implies continuous.
(C) A function is differentiable at a jump.
(D) A function is differentiable at a vertical tangent.

Solution
Answer: (B). If f is differentiable at c, it is continuous at c; the converse is false.

Question 9 (Free response) NO CALCULATOR

Use the graph of h above. List every x in $[-4, 6]$ where h is not differentiable, and give the reason for each.

Solution

$x = -4$: endpoint. $x = -2$: sharp corner. $x = 1$: hole (discontinuous). $x = 3$: vertical tangent. $x = 4$: jump (discontinuous). $x = 6$: endpoint.